Ibrahim Ali Khan

AI SOFTWARE ENGINEER

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PROFILE

A problem-solving software engineer with 3+ years of experience building full-stack platforms and intelligent automation tools, currently completing an M.Sc. in Computer Engineering at Paderborn University. Over time, my focus has expanded into AI engineering, where I work on LLM-powered features, AI agents, and systems that streamline complex workflows and reduce manual effort. As a Founding Engineer at Optimaite, I am helping build data-sovereign AI software for regulated, document-intensive environments. I enjoy turning complex problems into practical, reliable, and user-friendly solutions, with a strong focus on clean engineering, scalability, and continuous improvement. Available for full-time roles from November 2026.

EDUCATION

UNIVERSTÄT PADERBORN — Paderborn, Germany

Master of Science (M.Sc.), Computer Engineering — *March 2024 - Expected October 2026*

Focus: Web Security, Computer Architecture, Network Systems

Master's Thesis: *"AI-Assisted Hospital Management: A Modular, FHIR-Based System with Privacy-Preserving Design for the German Healthcare Context"*

- Designing a modular, privacy-preserving hospital management system with HL7 FHIR-native interoperability, role-differentiated access, and audit logging.
- Implementing GDPR-aware data handling in the German regulatory context (ePA), including patient-controlled data sharing and reversible pseudonymisation.
- Evaluating persona-specific AI agents for clinical workflows, benchmarking local vs. cloud LLMs on relevance, grounding, hallucination rate, latency, cost, and infrastructure requirements.

SIDDAGANGA INSTITUTE OF TECHNOLOGY (SIT) — Tumkur, India

Bachelor of Engineering (B.Eng.), Electronics & Communication — *July 2014 - August 2018*

Focus: Object-Oriented Programming, Statistical Computing, Embedded and Digital-system Design

EXPERIENCE

FRAUNHOFER IEM — Paderborn, Germany

Research Assistant/Founding Engineer — *November 2025 - Present*

Optimaite - AI Platform for Law Firms

- Founding Engineer at Optimaite, a Fraunhofer IEM GenAI Incubator spin-off, where I co-architected and shipped core capabilities for a data-sovereign AI platform that automates document-intensive workflows for law firms, supporting cloud and on-premise deployments.
- Built RAG pipelines and tool-using AI agents for legal and enterprise workflows, integrating embeddings, vector search, configurable retrieval, structured outputs, validation, human review, and failure handling.
- Implemented AI observability and tenant-aware security with prompt/retrieval/tool-call tracing, latency and failure monitoring, authorization, audit logging, and data-isolation controls for sensitive documents.
- Standardized containerized deployments with Docker and Coolify, reducing release time by almost 50% and improving deployment repeatability.

METALABS SERVICES LLP — Hyderabad, India

Software Developer — *April 2021 - November 2023*

WriteGenius AI - AI Content Platform

- Led a team of 4 developers to deliver an AI long-form content product, integrating ChatGPT with custom prompt pipelines and Python-based web scraping.
- Built the web interface with React, Tailwind CSS, and MongoDB, increasing platform engagement by 40% over 6 months.

TechStyleOS (BOND) - Fashion E-Commerce Platform

- Lead developer in an offshore team; migrated the BOND application from AngularJS to React, improving load time and performance drastically.
- Improved scalability through database splitting with SQL and Express; CI/CD with Docker, Jenkins, and Bitbucket.

Catalog Sync - Product Synchronization Platform

- Designed database schemas and GraphQL APIs; integrated WooCommerce and Shopify APIs to synchronize products between the platform and client stores.
- Built the UI with Vue, Quasar, and SCSS alongside app development with MongoDB, improving performance by 25%.

LineSheets Studio - B2B Fashion Catalog Tool

- Led the UI modernization of LineSheets Studio, improving usability and engagement through a redesigned Vue and Quasar interface.

LEARNYST INSIGHT PRIVATE LIMITED **Bengaluru, India**

Software Developer Intern *February 2019 - June 2019*

- Developed reusable web and mobile test-automation frameworks using Java, Selenium, Appium, Maven, JMeter, and Cucumber, reducing repetitive manual regression testing.
- Implemented automated functional, performance, and acceptance-testing workflows for web and mobile product releases.

TECHNICAL SKILLS

Applied AI: LLM Integration, RAG, AI Agents, Human-in-the-Loop Workflows, LLM Guardrails, Embeddings, Vector Databases, Prompt Engineering, LLM Evaluation, AI Observability, Model Benchmarking, Privacy-Preserving AI

Backend & APIs: Python, FastAPI, Node.js, Express, REST APIs, GraphQL, Microservices

Frontend: TypeScript, JavaScript, React, Next.js, Vue, Angular, Tailwind CSS, HTML, CSS

Data & Retrieval: SQL, MongoDB, Vector Databases, Database Modelling, Semantic Search, Data Pipelines

Cloud, DevOps & Delivery: Docker, Kubernetes, CI/CD, Azure DevOps, Jenkins, Coolify, Git, Bitbucket, On-Premise Deployment, Model Serving

Engineering Practices: System Design, Data Sovereignty, Technical Research, Agile Development

LANGUAGES

English (C1) · German (B1) · Urdu (C1) · Hindi (C1)

CERTIFICATIONS

Nagarro Career Bootcamp - Web Development **New Delhi, India**

Coding Blocks *May 2021*

Full Stack Development **Bangalore, India**

Aradhya Brilliance Center (ABC) *January 2020*

ACCOMPLISHMENTS

Spot Award - Top performer, Q2 **July 2023**

For TechStyleOS (BOND) at METALABS SERVICES LLP

Spot Award - Top performer, Q4 **July 2022**

For Catalog Sync at METALABS SERVICES LLP

Best Project Award - “Gesture Controlled Master and Slave Robot” **May 2016**

Technosphere Connect (Bachelors)